CYBERSECURITY RECONNAISSANCE FRAMEWORK

Attack Surface Discovery Tool

Modular multi-threaded Python framework for continuous external exposure mapping and vulnerability analysis

CLI Engine v1.0

Python 3.x / Concurrent Execution

EXECUTIVE OVERVIEW

Addressing Shadow IT & Unmanaged Exposures

Continuous Visibility

Automates mapping of public digital assets before adversarial targeting.

Concurrent Enumeration

High-speed multi-threaded workers analyze subdomains and services.

Proactive Defense

Flags missing security headers and exposed administrative interfaces.

SYSTEM ARCHITECTURE

4-Stage Reconnaissance Pipeline

01

DNS Resolution

Queries A, AAAA, MX, TXT, and NS infrastructure records.

02

Subdomain Brute

20-worker thread pool resolves wordlist targets to live IPs.

03

Port Probing

Scans common infrastructure and administrative ports.

04

Header Audit

Evaluates server banners and core HTTP security policies.

CAPABILITIES

Technical Scanning Capabilities

Multi-Threaded Execution

Concurrent standard sockets eliminate blocking latency during scanning.

Security Header Verification

Detects missing CSP, HSTS, X-Frame-Options, and XSS Protection headers.

Banner Fingerprinting

Extracts target server software versions and exposed HTTP headers.



MODULE DEEP DIVE

DNS & Subdomain Discovery Engine

DNS Record Harvesting

A / AAAA: Resolves IPv4 and IPv6 host mappings.

MX Records: Identifies primary mail server locations.

TXT Records: Extracts SPF, DKIM, and verification keys.

NS Records: Maps authoritative domain name servers.

Subdomain Brute Force

Wordlist Engine: Tests high-probability hosts (dev, api, vpn).

Socket Resolution: Standard OS lookups prevent library lag.

Active Asset List: Filters non-resolving endpoints automatically.

Thread Pool: Concurrently scales up to 20 active workers.

SERVICE DETECTION

Targeted Port Scanning & Service Probe

Web Infrastructure

Scans ports 80, 443, 8080, and 8443 for exposed web applications.

Remote Access

Probes ports 21 (FTP), 22 (SSH), and 23 (Telnet) for access points.

Databases & Mail

Identifies exposed MySQL (3306), Postgres (5432), and SMTP (25).

BENCHMARKS & PERFORMANCE

Execution Efficiency

20x

Execution speedup achieved via concurrent ThreadPoolExecutor workers

SOLUTION COMPARISON

Custom Tooling vs Legacy Scanners

Heavy Commercial Scanners

High network overhead and complex setups.

Prone to generic false-positive alerts.

Slower execution across targeted micro-scopes.

Python ASD Script

Zero Overhead: Lightweight, standalone Python implementation.

Targeted Scope: Direct focus on high-impact exposures.

DevSecOps Ready: Easy integration into CI/CD pipelines.

FUTURE ROADMAP

Feature Enhancement Pipeline

1

OSINT Integration

Incorporate Shodan, Censys, and SecurityTrails API sources.

2

Directory Fuzzing

Add path enumeration for hidden endpoints and exposed files.

3

Automated Reporting

Export scan results directly to JSON and HTML executive dashboards.

SUMMARY & NEXT STEPS

Secure Your External Perimeter

Deploy continuous discovery tools to gain complete visibility over your digital attack surface.

Questions & Technical Discussion

Image Sources



https://cdn.dribbble.com/userupload/48626659/file/56aa82e869d907b8ffcd0e9df21387ef.png?resize=752x&vertical=center

Source: dribbble.com